Screen-free game kit · Design; not tested in practice

Cool Route



Learning goal

Find shade and rest spots in an illustrative area; tell surface temperature, air temperature and felt comfort apart.

Materials

The card pages of this PDF (A4, 100% scale), scissors, pencils; box labels go on the table.

How to play



- No time limit; nobody is rushed.
- A wrong answer costs nothing: read why, then try again.
- A team that wants help turns over İKO's hint card.
- At the end, say what you learned in one sentence and choose the real-life task.

Task flow

Ages 7–9

İKO will cross the school yard on a hot afternoon. Find the shady spots and the resting place on the map.

1. Find 3 shady spots and 1 resting place on the map. (This map is illustrative.)
2. The slide in the sun may be very hot. What do we do?

Ages 10–12

We will choose a waiting place for İKO. Compare the sample measurement cards and justify your decision.

1. What does each card measure or describe?
2. Where would you put the waiting place? Choose the best reason.
3. "Does the hottest measured spot have to be the least comfortable spot?"



Real-life task: With your teacher's permission, choose a shady spot and a sunny spot in the school yard at the same time. Look without touching the surfaces; write how you feel separately.

If a badge is used, it only shows the task was finished; it is not proof of learning.

Cool Route · Ages 7–9 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 7–9 · Hint



Look next to the trees, the wall and the awning.

Ages 7–9 · Question card

The slide in the sun may be very hot. What do we do?

A. We look from a distance without touching and tell an adult.

B. We touch it with our hand to check.

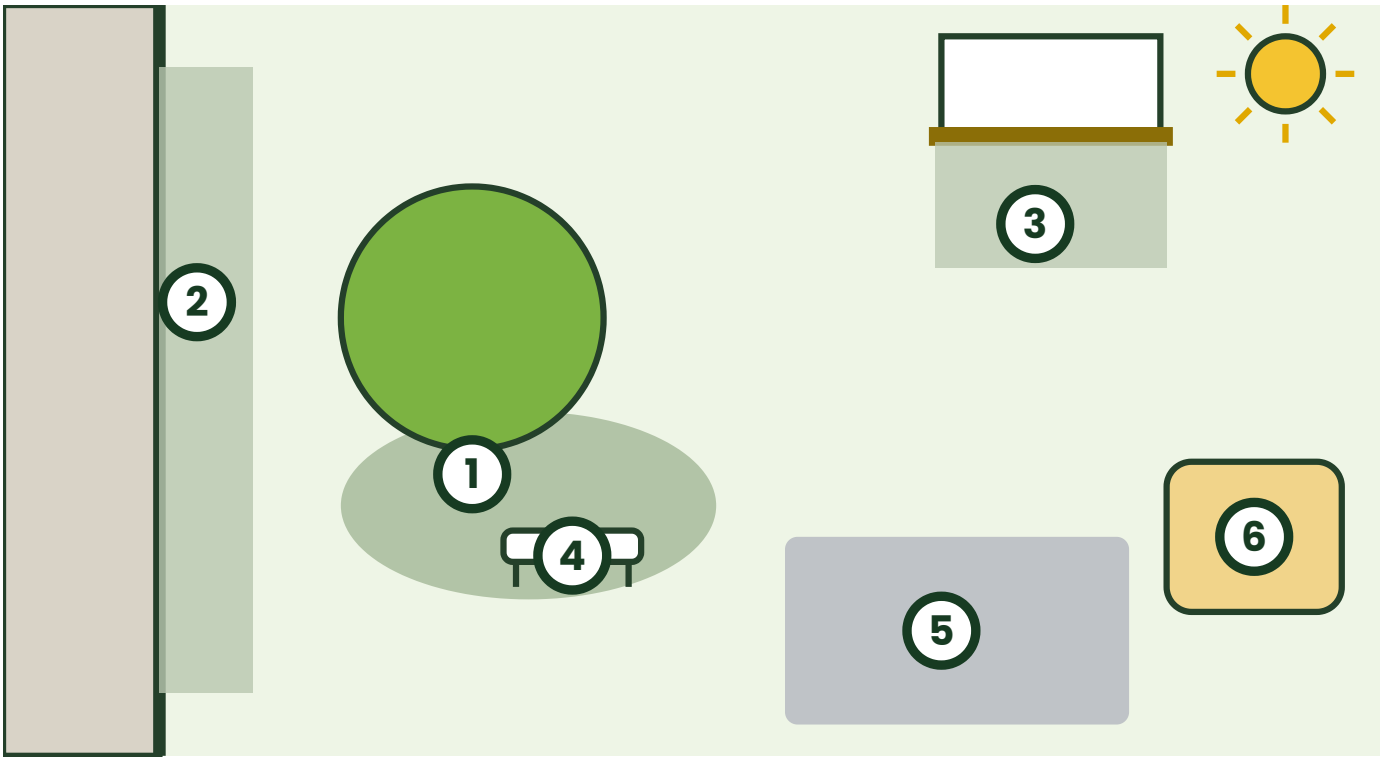
Ages 7–9 · Hint



Is it safe to test a hot surface with your hand?

Map (illustrative) · Ages 7–9

Find 3 shady spots and 1 resting place on the map. (This map is illustrative.)



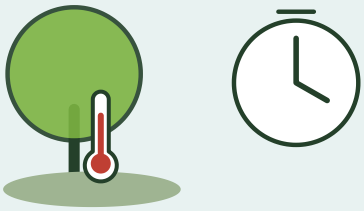
1. Under the tree
2. Foot of the wall
3. Under the awning
4. Bench next to the tree
5. Open asphalt area
6. Sandpit in the open

Map description: the sun is at the top right, in the afternoon. On the left is the school building wall. In the middle there is a big tree with a bench next to it. At the top right there is a door with an awning. At the bottom right there is an open asphalt area and a sandpit.

Cool Route · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm



Surface temperature



Air temperature



Felt comfort

Ages 10–12 · Cut-out cards

Infrared thermometer aimed at asphalt in the sun: 52 °C (SAMPLE)

Ages 10–12 · Cut-out cards

The same thermometer aimed at soil in tree shade: 36 °C (SAMPLE)

Ages 10–12 · Cut-out cards

Thermometer left in the shade for 3 minutes: 31 °C (SAMPLE)

Cool Route · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 10–12 · Cut-out cards

"I felt comfortable sitting on the bench."

Ages 10–12 · Cut-out cards

"My head felt hot standing in the sun."

Ages 10–12 · Hint



Was it aimed at a surface, measured in the air in the shade, or is it a person's feeling?

Ages 10–12 · Question card

Where would you put the waiting place? Choose the best reason.

A. In the tree shade: the sample surface reading is lower there and participants said they felt more comfortable; we should measure again at the same time in the real area.

B. Next to the asphalt: air temperature is the same everywhere.

C. In the tree shade: because it is definitely 16 degrees cooler.

Ages 10–12 · Hint



A good reason uses both the measurement and the feeling separately and states the limit of the sample data.

Ages 10–12 · Question card

"Does the hottest measured spot have to be the least comfortable spot?"

A. No; measurement and feeling are recorded separately.

B. Yes, always.

Cool Route · Ages 10–12 · ✂ Cut along the dashed line · card 95 × 86 mm

Ages 10–12 · Hint



Wind, humidity, time in shade and the person also affect the feeling.

Answers and explanations (teacher only)

Ages 7–9

1.1	Under the tree	Fits	The tree makes shade.
1.2	Foot of the wall	Fits	The wall blocks the sun.
1.3	Under the awning	Fits	The awning makes shade.
1.4	Bench next to the tree	Fits	We can sit and rest in the shade.
1.5	Open asphalt area	Does not fit	This is in the sun; no shade.
1.6	Sandpit in the open	Does not fit	This is in the sun; no shade.
2A	We look from a distance without touching and tell an adult.	Fits	Hot surfaces are not touched.
2B	We touch it with our hand to check.	Does not fit	Our hand could get burned; touching is not safe.

Ages 10–12

1	Infrared thermometer aimed at asphalt in the sun: 52 °C (SAMPLE)	Surface temperature	A reading aimed at a surface is surface temperature.
1	The same thermometer aimed at soil in tree shade: 36 °C (SAMPLE)	Surface temperature	Again a surface.
1	Thermometer left in the shade for 3 minutes: 31 °C (SAMPLE)	Air temperature	A thermometer left in the shade measures the air.
1	“I felt comfortable sitting on the bench.”	Felt comfort	This is a feeling; it varies from person to person.
1	“My head felt hot standing in the sun.”	Felt comfort	This is also a feeling, not a measurement.
2A	In the tree shade: the sample surface reading is lower there and participants said they felt more comfortable; we should measure again at the same time in the real area.	Fits	It separates measurement and feeling, and states the limit.
2B	Next to the asphalt: air temperature is the same everywhere.	Does not fit	We did not measure that; surface heat can also affect nearby conditions.
2C	In the tree shade: because it is definitely 16 degrees cooler.	Does not fit	52 and 36 are sample surface values, not an air temperature difference; it cannot be called “definite”.
3A	No; measurement and feeling are recorded separately.	Fits	Right. Surface and air temperature also differ (K13).

3B	Yes, always.	Does not fit	Feeling varies from person to person; a measurement alone does not tell comfort.

Cool Route: Shade can help surfaces heat up less (K13). Surface temperature, air temperature and how we feel are different things; they are recorded separately.